Trinh Thi Thuy Duong

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EDUCATION

Hanoi University of Science and Technology (HUST) — **Ha Noi, Viet Nam**
B.Sc. in Information Technology (English-taught) • CPA: 3.6/4.0 — *09/2022 – 09/2026 (Expected)*

- **Selected Coursework:** Machine Learning & Data Mining, Intro to AI, Data Structures & Algorithms, Applied Algorithms, Discrete Math, Probability & Statistics, Operating Systems.

Bac Ninh High School for the Talented — **Bac Ninh, Viet Nam**
Specialized Program in Literature • GPA: 9.1/10 — *09/2019 – 05/2022*

RESEARCH INTERESTS

- **Human–Agent Interaction:** Exploring and improving human–agent interaction across diverse contexts, with a particular focus on the quality of dialogue-based interaction.
- **Computational Sociolinguistics & Language Technology:** Investigating NLP problems in interdisciplinary, multilingual, and multicultural contexts, and how language models encode and propagate linguistic and cultural variation.

PUBLICATIONS

- **Thi-Thuy-Duong Trinh**, Ngoc-Hai Truong*, Trong-Huy Nguyen, and Thanh-Huong Le*. *Contrastive Preference Optimization for Low-Resource Vietnamese to Khmer Neural Machine Translation.* In Proceedings of the 14th International Symposium on Information and Communication Technology (SOICT '25), ACM, 2025. [DOI]

EXPERIENCE

Viettel Cyber Security, Viettel Group — **Ha Noi, Viet Nam**
AI Engineer (Promoted prior to graduation) — *03/2026 – Present*

- **Human–Agent Collaboration Platform (Internal Pilot):** Designed the orchestration architecture for a SOC coworking platform where human analysts and AI agents work as equal partners, supporting two-way task assignment and peer review.
- **RINA Expert (*Best Innovation Idea of the Year*):** Fine-tuned Small Language Models (SLMs) for threat-hunting tasks, enabling lightweight and secure local execution in air-gapped environments.

Viettel Cyber Security, Viettel Group — **Ha Noi, Viet Nam**
AI Engineer Intern — *03/2025 – 02/2026*

- **EDR-SLM Framework:** Fine-tuned SLMs (Qwen 0.5B–8B) via Unsloth and LoRA to replace heavy LLMs in multi-agent pipelines, optimizing execution outcomes, inference latency, and VRAM footprint for NL-to-SQL translation.
- **Self-Defending AI Agent Architecture:** Researched multi-turn adversarial exploits (prompt injection, logic deception) to design safety-aligned agent architectures capable of detecting malicious intents and enforcing safe execution policy during code/Bash generation.

BKAI NLP Lab (Hanoi University of Science and Technology) — **Ha Noi, Viet Nam**
Research Member • Supervisor: Assoc. Prof. Le Thanh Huong — *09/2024 – Present*

- **Graduation Thesis – Multi-Level Knowledge Distillation for Low-Resource NMT** (Grade: 9.3/10)
  - Designed a multi-level KD framework transferring capabilities from large teachers to compact students (Qwen3-1.7B, Qwen2.5-0.5B) via distribution-aware logit distillation, full-layer feature alignment, and intra-student cross-lingual contrastive learning.
  - Resolved teacher–student vocabulary mismatch using bipartite-graph span alignment; boosted 1.7B student chrF++ on FLORES-200 from 10.09 to 30.41 (Vi–Khm) and 6.54 to 30.86 (Vi–Lao).
- **Vietnamese–Khmer NMT Optimization** [GitHub]
  - Developed a 3-stage pipeline (Monolingual CPT → LoRA SFT → CPO) using 27K self-constructed preference triplets; surpassed Sailor2-1B, DPO, and GPT-4o-mini across BLEU, METEOR, COMET, and COMET-KIWI.

AWARDS AND CERTIFICATIONS

- **Outstanding Intern of the Year**, Viettel Cyber Security (12/2025) — Received early full-time offer before graduation.
- **Best Innovation Idea of the Year**, Viettel Cyber Security (06/2026) — Awarded for *RINA Expert* SLM threat-hunting framework.
- **Second Prize**, Viettel AI Race 2025 (Natural Language Processing Track) — Ranked 2nd out of 600+ participating teams nationwide.
- **IELTS Academic 6.5** — British Council.
- **IBM Data Science Professional Certificate** — Coursera.

TECHNICAL SKILLS

- **Languages:** Python, C/C++, SQL, Bash
- **Frameworks & ML:** PyTorch, HuggingFace, Unsloth, LoRA/PEFT, LangChain
- **Tools:** Docker; Git; Langfuse, Grafana; Qdrant, Milvus (vector database)

EXTRACURRICULAR ACTIVITIES

- **Club President**, Hai Au Voluntary Club {2019 – 2021}: Spearheaded relief and fundraising campaigns during COVID-19 and natural disasters in Central Vietnam.
- **Co-Founder & Head of Content**, LaCul Project {2020 – 2021}: Led educational initiatives exploring global languages and cultures through published media.